

HW1 - STAT 701

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Before Reading:

To make the proof more readable, I will miss or gap some natural or not important facts or notations during my writing. If you feel it hard to see, you can refer the appendix after the proof, where I will try to explain some simple conclusions (will be marked) more clearly. In case that you misunderstand the mark, I will add the mark just after those formulas between \$ and before those between \$\$.

And I have to claim that the appendix is of course a part of my assignment, so the reference of it is required. Enjoy your grading!

Problem 1. Ex. 1.3.

Suppose that $m_t = c_0 + c_1 t + c_2 t^2$, $t = 0, \pm 1, \dots$,

(a) Show that

$$m_t = \sum_{i=-2}^2 a_i m_{t+i} = \sum_{i=-3}^3 b_i m_{t+i}, \quad t = 0, \pm 1, \dots$$

where $a_2 = a_{-2} = -\frac{3}{35}$, $a_1 = a_{-1} = \frac{12}{35}$, $a_0 = \frac{17}{35}$, and $b_3 = b_{-3} = -\frac{2}{21}$, $b_2 = b_{-2} = \frac{3}{21}$, $b_1 = b_{-1} = \frac{6}{21}$, $b_0 = \frac{7}{21}$.

(b) Suppose that $X_t = m_t + Z_t$, where $(Z_t)_{t \in \mathbb{Z}}$ is an independent sequence of normal random variables, each with mean 0 and variance σ^2 . Let

$$U_t = \sum_{i=-2}^2 a_i X_{t+i} \quad \text{and} \quad V_t = \sum_{i=-3}^3 b_i X_{t+i}$$

then

- (i) Find the means and variances of U_t and V_t .
- (ii) Find the correlations between U_t and U_{t+1} and between V_t and V_{t+1} .
- (iii) Which of the two filtered series $(U_t)_{t \in \mathbb{Z}}$ and $(V_t)_{t \in \mathbb{Z}}$ would you expect to be smoother in appearance?

Sol.

(a) Compare the coefficient of c_i for $i = 0, 1, 2$ respectively, we get

$$a_{-2} + a_{-1} + a_0 + a_1 + a_2 = 1,$$

$$(t-2)a_{-2} + (t-1)a_{-1} + ta_0 + (t+1)a_1 + (t+2)a_2 = t,$$

$$(t-2)^2 a_{-2} + (t-1)^2 a_{-1} + t^2 a_0 + (t+1)^2 a_1 + (t+2)^2 a_2 = t^2,$$

and

$$b_{-3} + b_{-2} + b_{-1} + b_0 + b_1 + b_2 + b_3 = 1,$$

$$(t-3)b_{-3} + (t-2)b_{-2} + (t-1)b_{-1} + tb_0 + (t+1)b_1 + (t+2)b_2 + (t+3)b_3 = t$$

$$(t-3)^2b_{-3} + (t-2)^2b_{-2} + (t-1)^2b_{-1} + t^2b_0 + (t+1)^2b_1 + (t+2)^2b_2 + (t+3)^2b_3 = t^2$$

if and only if

$$m_t = \sum_{i=-2}^2 a_i m_{t+i} = \sum_{i=-3}^3 b_i m_{t+i}, \quad t = 0, \pm 1, \dots$$

Simplify the two systems above, we obtained

$$a_{-2} + a_{-1} + a_0 + a_1 + a_2 = 1,$$

$$-2a_{-2} + -a_{-1} + a_1 + 2a_2 = t,$$

$$4a_{-2} + 1a_{-1} + a_0 + 1a_1 + 2a_2 = t^2,$$

and

$$b_{-3} + b_{-2} + b_{-1} + b_0 + b_1 + b_2 + b_3 = 1,$$

$$-3b_{-3} + -2b_{-2} + -b_{-1} + b_1 + 2b_2 + 3b_3 = t,$$

$$9b_{-3} + 4b_{-2} + b_{-1} + b_1 + 4b_2 + 9b_3 = t^2$$

equivalently. Substituting $a_2 = a_{-2} = -\frac{3}{35}$, $a_1 = a_{-1} = \frac{12}{35}$, $a_0 = \frac{17}{35}$, and $b_3 = b_{-3} = -\frac{2}{21}$, $b_2 = b_{-2} = \frac{3}{21}$, $b_1 = b_{-1} = \frac{6}{21}$, $b_0 = \frac{7}{21}$ will verify the simplified systems above and we are done.

(b) (i) We know

$$\mathbb{E}U_t = \sum_{i=-2}^2 a_i \mathbb{E}(m_{t+i} + Z_{t+i}) = m_t$$

$$\mathbb{E}V_t = \sum_{i=-3}^3 b_i \mathbb{E}(m_{t+i} + Z_{t+i}) = m_t$$

and

$$\begin{aligned} \text{Var}U_t &= \mathbb{E} \left(\sum_{i=-2}^2 a_i (m_{t+i} + Z_{t+i}) - m_t \right)^2 \\ &= \mathbb{E} \left(\sum_{i=-2}^2 a_i Z_{t+i} \right)^2 = \sum_{i=-2}^2 a_i^2 \sigma^2 = \sigma^2 \\ \text{Var}V_t &= \mathbb{E} \left(\sum_{i=-3}^3 b_i (m_{t+i} + Z_{t+i}) - m_t \right)^2 \\ &= \mathbb{E} \left(\sum_{i=-3}^3 b_i Z_{t+i} \right)^2 = \sum_{i=-3}^3 b_i^2 \sigma^2 = \sigma^2 \end{aligned}$$

(ii) Notice that

$$U_t - \mathbb{E}U_t = \sum_{i=-2}^2 a_i Z_{t+i}, \quad V_t - \mathbb{E}V_t = \sum_{i=-3}^3 b_i Z_{t+i},$$

and hence

$$\begin{aligned}
\text{Cov}(U_t, U_{t-1}) &= \mathbb{E} \left(\sum_{i=-2}^2 a_i Z_{t+i} \right) \left(\sum_{i=-2}^2 a_i Z_{t+1+i} \right) \\
&= a_{-1}a_{-2} + a_0a_{-1} + a_1a_0 + a_2a_1 = \frac{48}{165} \\
\text{Cov}(V_t, V_{t-1}) &= \mathbb{E} \left(\sum_{i=-3}^3 b_i Z_{t+i} \right) \left(\sum_{i=-3}^3 b_i Z_{t+1+i} \right) \\
&= b_{-2}b_{-3} + b_{-1}b_{-2} + b_0b_{-1} + b_1b_0 + b_2b_1 + b_3b_2 = \frac{12}{49}
\end{aligned}$$

(iii) The covariance of V_t, V_{t+1} is smaller than that of U_t, U_{t+1} , and since V_t, U_t have equal variances, we may know that U_t, U_{t+1} are more dependent with each other than V_t, V_{t+1} , so $(U_t)_{t \in \mathbb{Z}}$ will be smoother.

Problem 2

Suppose that $m_t = c_0 + c_1t + c_2t^2, t = 0, \pm 1, \dots$,

(a) Show that

$$m_t = \sum_{i=-2}^2 a_i m_{t+i} = \sum_{i=-3}^3 b_i m_{t+i}, \quad t = 0, \pm 1, \dots$$

where $a_2 = a_{-2} = -\frac{3}{35}, a_1 = a_{-1} = \frac{12}{35}, a_0 = \frac{17}{35}$, and $b_3 = b_{-3} = -\frac{2}{21}, b_2 = b_{-2} = \frac{3}{21}, b_1 = b_{-1} = \frac{6}{21}, b_0 = \frac{7}{21}$.

(b) Suppose that $X_t = m_t + Z_t$, where $(Z_t)_{t \in \mathbb{Z}}$ is an independent sequence of normal random variables, each with mean 0 and variance σ^2 . Let

$$U_t = \sum_{i=-2}^2 a_i X_{t+i} \quad \text{and} \quad V_t = \sum_{i=-3}^3 b_i X_{t+i}$$

then

- Find the means and variances of U_t and V_t .
- Find the correlations between U_t and U_{t+1} and between V_t and V_{t+1} .
- Which of the two filtered series $(U_t)_{t \in \mathbb{Z}}$ and $(V_t)_{t \in \mathbb{Z}}$ would you expect to be smoother in appearance?

Sol.

(a) Compare the coefficient of c_i for $i = 0, 1, 2$ respectively, we get

$$\begin{aligned}
a_{-2} + a_{-1} + a_0 + a_1 + a_2 &= 1, \\
(t-2)a_{-2} + (t-1)a_{-1} + ta_0 + (t+1)a_1 + (t+2)a_2 &= t, \\
(t-2)^2a_{-2} + (t-1)^2a_{-1} + t^2a_0 + (t+1)^2a_1 + (t+2)^2a_2 &= t^2,
\end{aligned}$$

and

$$\begin{aligned}
b_{-3} + b_{-2} + b_{-1} + b_0 + b_1 + b_2 + b_3 &= 1, \\
(t-3)b_{-3} + (t-2)b_{-2} + (t-1)b_{-1} + tb_0 + (t+1)b_1 + (t+2)b_2 + (t+3)b_3 &= t \\
(t-3)^2b_{-3} + (t-2)^2b_{-2} + (t-1)^2b_{-1} + t^2b_0 + (t+1)^2b_1 + (t+2)^2b_2 + (t+3)^2b_3 &= t^2
\end{aligned}$$

if and only if

$$m_t = \sum_{i=-2}^2 a_i m_{t+i} = \sum_{i=-3}^3 b_i m_{t+i}, \quad t = 0, \pm 1, \dots$$

Simplify the two systems above, we obtained

$$\begin{aligned} a_{-2} + a_{-1} + a_0 + a_1 + a_2 &= 1, \\ -2a_{-2} + -a_{-1} + a_1 + 2a_2 &= t, \\ 4a_{-2} + 1a_{-1} + a_0 + 1a_1 + 2a_2 &= t^2, \end{aligned}$$

and

$$\begin{aligned} b_{-3} + b_{-2} + b_{-1} + b_0 + b_1 + b_2 + b_3 &= 1, \\ -3b_{-3} + -2b_{-2} + -b_{-1} + b_1 + 2b_2 + 3b_3 &= t, \\ 9b_{-3} + 4b_{-2} + b_{-1} + b_1 + 4b_2 + 9b_3 &= t^2 \end{aligned}$$

equivalently. Substituting $a_2 = a_{-2} = -\frac{3}{35}$, $a_1 = a_{-1} = \frac{12}{35}$, $a_0 = \frac{17}{35}$, and $b_3 = b_{-3} = -\frac{2}{21}$, $b_2 = b_{-2} = \frac{3}{21}$, $b_1 = b_{-1} = \frac{6}{21}$, $b_0 = \frac{7}{21}$ will verify the simplified systems above and we are done.

(b) (i) We know

$$\begin{aligned} \mathbb{E}U_t &= \sum_{i=-2}^2 a_i \mathbb{E}(m_{t+i} + Z_{t+i}) = m_t \\ \mathbb{E}V_t &= \sum_{i=-3}^3 b_i \mathbb{E}(m_{t+i} + Z_{t+i}) = m_t \end{aligned}$$

and

$$\begin{aligned} \text{Var}U_t &= \mathbb{E} \left(\sum_{i=-2}^2 a_i (m_{t+i} + Z_{t+i}) - m_t \right)^2 \\ &= \mathbb{E} \left(\sum_{i=-2}^2 a_i Z_{t+i} \right)^2 = \sum_{i=-2}^2 a_i^2 \sigma^2 = \sigma^2 \\ \text{Var}V_t &= \mathbb{E} \left(\sum_{i=-3}^3 b_i (m_{t+i} + Z_{t+i}) - m_t \right)^2 \\ &= \mathbb{E} \left(\sum_{i=-3}^3 b_i Z_{t+i} \right)^2 = \sum_{i=-3}^3 b_i^2 \sigma^2 = \sigma^2 \end{aligned}$$

(ii) Notice that

$$U_t - \mathbb{E}U_t = \sum_{i=-2}^2 a_i Z_{t+i}, \quad V_t - \mathbb{E}V_t = \sum_{i=-3}^3 b_i Z_{t+i},$$

and hence

$$\begin{aligned} \text{Cov}(U_t, U_{t-1}) &= \mathbb{E} \left(\sum_{i=-2}^2 a_i Z_{t+i} \right) \left(\sum_{i=-2}^2 a_i Z_{t+1+i} \right) \\ &= a_{-1}a_{-2} + a_0a_{-1} + a_1a_0 + a_2a_1 = \frac{48}{165} \\ \text{Cov}(V_t, V_{t-1}) &= \mathbb{E} \left(\sum_{i=-3}^3 b_i Z_{t+i} \right) \left(\sum_{i=-3}^3 b_i Z_{t+1+i} \right) \\ &= b_{-2}b_{-3} + b_{-1}b_{-2} + b_0b_{-1} + b_1b_0 + b_2b_1 + b_3b_2 = \frac{12}{49} \end{aligned}$$

(iii) The covariance of V_t, V_{t+1} is smaller than that of U_t, U_{t+1} , and since V_t, U_t have equal variances, we may know that U_t, U_{t+1} are more dependent with each other than V_t, V_{t+1} , so $(U_t)_{t \in \mathbb{Z}}$ will be smoother.